

Duc Duong

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EDUCATION

Grinnell College

Grinnell, IA

B.A. in Computer Science & Mathematics (Concentration: Statistics)

Aug. 2024 – May 2028

- **GPA:** 3.8/4.0 - Dean's List 2024, 2025
- **Courses:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, Machine Learning, Artificial Intelligence, Probability & Statistics, Linear Algebra

SKILLS SUMMARY

Languages: Python, C++, C, Java, R, MATLAB, Bash, Git, LaTeX

ML/AI Frameworks: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Fine-Tuning, LoRA/QLoRA, RL, NLP

LLM Inference: KV-Cache Optimization, FlashAttention-2, Quantization, RAG, Distributed Training (FSDP/DDP)

Data & Cloud: SQL, Docker, Kubernetes, AWS, GCP, MLflow, Linux, CI/CD

PUBLICATIONS

Duc Duong, Hoang Anh Duy Le, Jianwen Xie, Anshumali Shrivastava, Zhaozhuo Xu, “Forget Without Compromise: Nexus Sampling for Streaming KV-Cache Eviction Under Fixed Budgets,” *arXiv:2606.23961*.

EXPERIENCE & LEADERSHIP

Stevens Institute of Technology | AI Research Intern

Jun 2026 – Present

- **Context Management.** Research **efficient context management** for long-context and agentic LLMs, focusing on KV-cache eviction and memory under fixed budgets.
- **LLM Inference.** Authored *Nexus Sampling*, a training-free eviction method matching dense attention within 1 point on **LongBench** at 80% eviction while cutting cache memory 5–10× at 1.22–1.30× decode throughput.
- **Agentic RL.** Investigating context management as a next direction for long-running agents, where retention and memory policies shape **RL** over multi-turn traces using **Apex Agent** as a benchmark.

Gtel Data Research Group | R&D Software Engineering Intern

May 2025 - Aug 2025

- **Model Fine-Tuning.** Fine-tuned an Ancient-to-Modern Vietnamese translation model with LoRA/QLoRA, reaching a 49 BLEU score and a 15-point gain over the base model.
- **Data Engineering.** Engineered a 1M+ sample synthetic dataset via a **round-robin algorithm** across Gemini API keys, enabling high-throughput data curation.
- **Collaboration.** Presented **ML topics** (Transformers, CNNs, Diffusion) weekly to senior engineers, enhancing communication and coaching skills.

Data Glacier | Data Science Intern

Feb. 2025 - May 2025

- **Model Development.** Built a hate-speech detection engine (BERT + Hugging Face Transformers) reaching 0.92 F1 through custom fine-tuning of 50,000+ documents.
- **Inference Optimization.** Cut inference latency 45% with Hugging Face Accelerate and ONNX Runtime, deploying containerized microservices on **Docker** + **GCP** with **CI/CD**.
- **System Monitoring.** Authored 10+ technical reports diagnosing pipeline bottlenecks, reducing system errors 25% and strengthening reproducible NLP research.

Vietnamese Student Association @ Grinnell College | Executive, VSA Cabinets

Jan. 2025 - Present

- **Leadership Development.** Coordinated 8-member board to deliver 6+ cultural events for 300+ attendees
- **Project Management.** Managed \$5k+ in logistics, ensuring accountability in budgeting and decision-making.

AWARDS AND CERTIFICATIONS

Outstanding Award, SCUDEM Mathematical Modeling: AI Ouroboros (SIMIODE, 2026)

First Prize, USPROC Statistics Research Competition (ASA & CAUSE, 2025)

Iowa Collegiate Mathematics Competition: 1st Place, 96/100 (2026); 2nd Place, 99/100 (2025)

Certifications: Oracle Certified AI Foundations Associate; Oracle Certified Data Science Professional (2025); Claude Code in Action (Anthropic); UR2PhD Pre-Research Experience (CRA, 2025)